UNRELATED DONOR BONE MARROW TRANSPLANTATION IN CHILDREN:

SURVIVAL ANND KEY DETERMINANTS

Abstract

In this retrospective analysis of 187 pediatric patients who underwent unmanipulated allogeneic unrelated donor hematopoietic stem cell transplantation, we investigated key factors influencing post-transplant survival outcomes. Using a comprehensive statistical approach combining Kaplan-Meier analysis for initial exploration, parametric model selection, and LASSO-guided semi-parametric Cox proportional hazards modeling for detailed investigation, we identified several significant predictors of survival, providing valuable insights for improving risk assessment and treatment strategies in pediatric bone marrow transplantation.

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Bone marrow donation involves extracting bone marrow from the hip bone to collect stem cells, which can develop into red blood cells, white blood cells, and platelets to save patients' lives.

INTRODUCTION

Allogeneic unrelated donor hematopoietic stem cell transplantation (allo-HSCT) represents a crucial therapeutic option for children with various hematologic diseases. Understanding the factors that influence survival outcomes is essential for improving treatment strategies and patient care.

DATA

The Bone Marrow Transplant: Children dataset available on the UCI Machine Learning Repository focuses on clinical outcomes for pediatric patients who have undergone bone marrow transplantation. The dataset comprises records from 187 pediatric patients who underwent unmanipulated allogeneic unrelated donor hematopoietic stem cell transplantation between 2000 and 2016, containing comprehensive information about patient characteristics, treatment details, and outcomes. This valuable collection includes key variables such as patient demographics, disease types (including ALL and AML), donor compatibility metrics, treatment protocols, and survival outcomes, making it particularly useful for analyzing factors that influence transplant success in pediatric populations.

OBJECTIVE

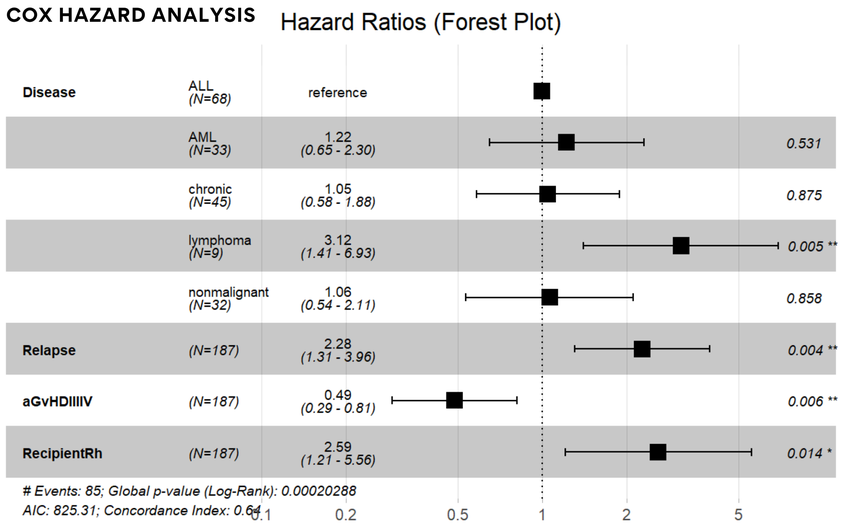
To identify and analyze key factors affecting survival outcomes in pediatric patients undergoing bone marrow transplantation, including:
Patient demographic characteristics, Disease-specific factors, Donor compatibility metrics, Treatment protocols.

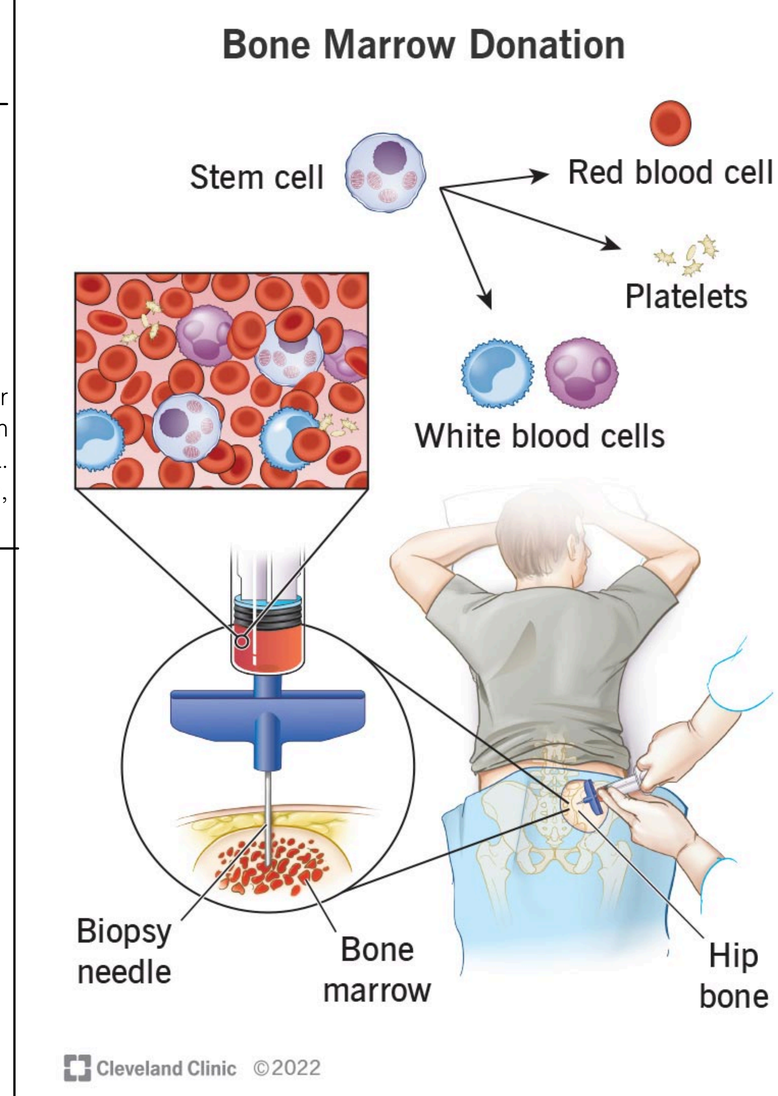
ANALYISIS

Initial Survival Analysis:
Kaplan-Meier curves were utilized to identify subgroups with the most pronounced survival differences, providing initial insights into potential prognostic factors. The parametric models are constructed based on graphical comparison and selected by the Likelihood Ratio Test.

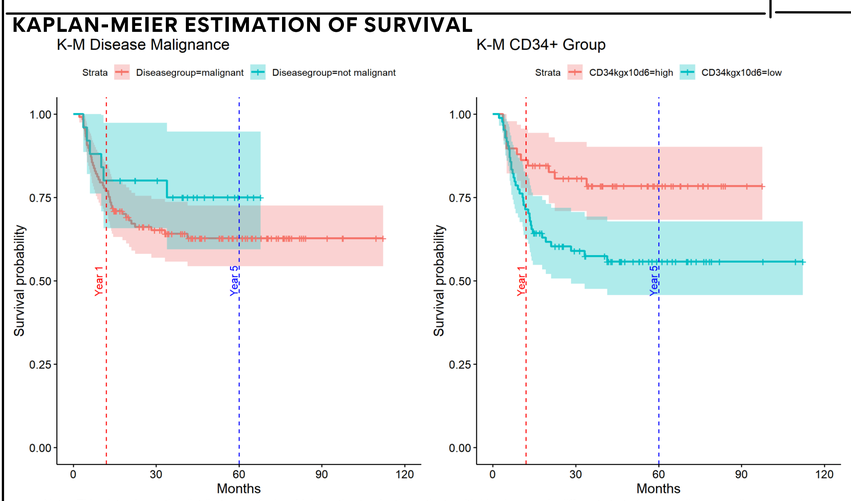
Variable Selection and Multivariate Analysis:
LASSO regression combined with univariate Cox models was employed for systematic variable selection. Selected significant variables were then incorporated into a comprehensive multivariate Cox proportional hazards model. Proportional hazards assumptions were verified and satisfied for all variables, eliminating the need for stratification.

COX HAZARD ANALYSIS

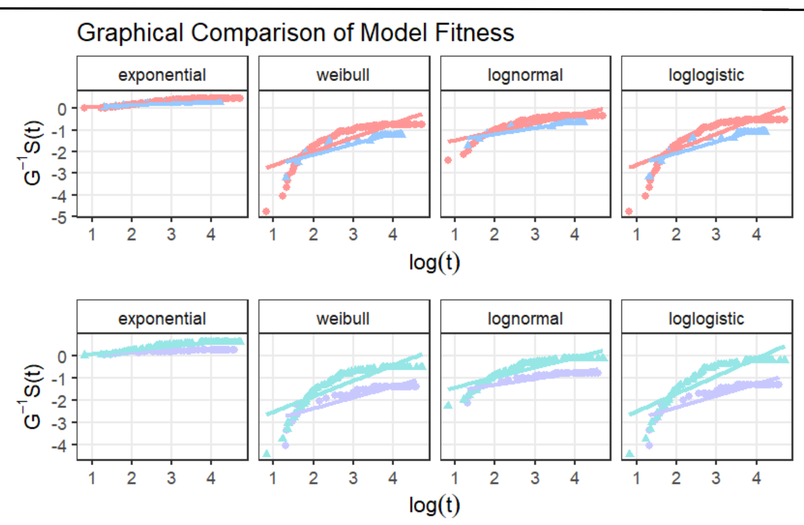




KAPLAN-MEIER ESTIMATION OF SURVIVAL



Notes:This plots shows the distribution of Kaplan–Meier estimates for the patients with blood cancer in different strata (groups) disease malignance and groups of CD34+ cells in the range of about 120 months (10 years).



Notes: This plot shows the fitness of potential parametric models to the data on the malignance (upper figure) and CD34+ group (lower figure). The parallel lines with most points are aligned with the lines would indicate a comparative good fit, which is observed to be the lognormal model.

LRT SUMMERIES OF PARAMETRIC MODELS

Terms	-2*LL	Df	Pr(>Chi)	CompareTo
Malignance	522.536	NA	NA	NA
Malignance + CD34+	516.450	1	0.014	Malignance
Malignance*CD34	516.333	1	0.732	Malignance+CD34+

Note:
This table shows the Likelihood Ratio Test (LRT) results including the test statistics $-2 * LL$, the degrees of freedom Df and the corresponding p-values $Pr(> Chi)$ under chi-square distributions for selecting the parametric model. In this table, the model 3 with the interaction term also includes the main effects from previous model. The result shows that the second model is preferred compared to the third and first model.

THE SELECTED LOGNORMAL PARAMETRIC MODEL

Term	Value	Std. Error	z	p
(Intercept)	8.774	0.494	17.76	< 2e-16
not malignant	0.081	0.638	0.13	0.899
low CD34+	-1.186	0.496	-2.39	0.017
Log(scale)	0.753	0.119	6.33	2.5e-10

Note:
This table shows the summay of estimates and their standard errors from the lognormal parametric model for predicting the survival time of patients. The z (z-score) and p (p-value) here are referred for any Wald Tests conducted on the coefficients to see if each covariate has influence on the expected survival time.

RESULTS/FINDINGS

Survival by Disease Type (Kaplan-Meier Analysis):

- Patients with nonmalignant diseases demonstrated significantly higher survival probabilities compared to those with malignant conditions.
- Distinct survival differences emerged at Year 1 and Year 5, highlighting the impact of disease type on outcomes.

Impact of CD34+ Cell Dosage:

- Patients receiving high CD34+ cell doses had notably longer survival times than those with low doses, emphasizing the significance of cell dosage in transplantation success.

Key Predictors of Survival (Multivariate Cox Analysis):

- Lymphoma patients had a significantly higher mortality risk compared to ALL patients (HR = 3.12, p = 0.005).
- Disease relapse strongly predicted adverse outcomes (HR = 2.28, p = 0.004).
- Acute GvHD (grades III/IV) showed a protective effect (HR = 0.49, p = 0.006).
- The Cox model achieved statistical significance (Global p-value = 0.0002) with a concordance index of 0.64.

Parametric Model Fit:

- Lognormal distribution provided the best fit for survival data, both by malignance status and CD34+ levels, enabling more accurate predictions for future patients.

CONCLUSION

Our analysis revealed critical insights for improving pediatric bone marrow transplantation outcomes. The findings emphasize that disease type significantly impacts survival, with lymphoma patients requiring particularly careful monitoring and potentially modified treatment approaches compared to other conditions. Disease relapse emerged as a crucial turning point in patient outcomes, highlighting the vital importance of post-transplant vigilance and early intervention strategies. Interestingly, the development of acute GvHD showed an unexpected protective effect, suggesting potential benefits that need to be balanced against its known risks. The superior outcomes observed in patients with nonmalignant diseases and those receiving higher CD34+ cell doses provide valuable guidance for treatment optimization. These findings collectively offer a foundation for more personalized treatment strategies, enabling clinicians to better assess risks and tailor their approach based on individual patient characteristics, ultimately working toward improved survival rates in pediatric bone marrow transplantation.